Electricity in the home

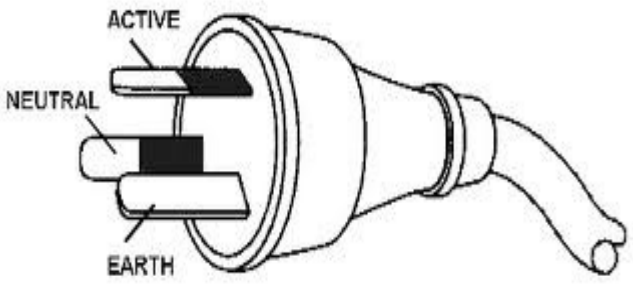
A battery produces direct current (_ _) while the electricity to supplied to our homes (called mains electricity) is alternating current or _ _.

In direct current the electrons flow from the _____(-) terminal to the _____ (+) terminal. With AC the electrons jiggle back and forwards through the wire. Each jiggling electron passes its energy onto the _____ electron.

The elelctricity in our home is a large _____ circuit. Each appliance is on its own _____ with its own _____.

The electrical cables are made of _____ separate wires.

3 pin plug



Draw the diagram and colour each pin the appropriate colour.

The live wire is called _____ and is coloured ___. The neutral wire is coloured _____.

The earth wire is a safety feature it is usually _____ or _____.

Electricity supplied to homes in Australia is supplied at _____V which is enough to kill a person.

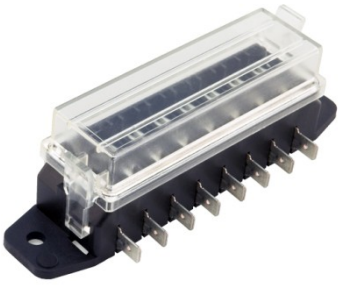
In case of an accident the electricity will flow through the _____ wire instead of an appliance or a person.

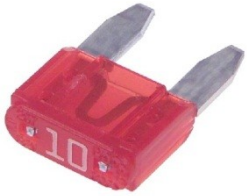
Electrical Safety

When electricity flows through a wire it gets _____. If the wires got too hot they could _____ their _____ coating or catch _____.

Appliances that are not working properly can cause too much _____ to flow through _____ _____ which can also cause a fire.

Fuses & Circuit Breakers



Fuses have a higher _____ with a lower _____ _____ than the other _____ wiring in the house. If too much _____ flows the wire in the fuse will _____ before any of the other wiring. Once the wire in the fuse melts the _____ is _____ and no electricity will flow. We say the fuse has _____.

When a fuse blows you should check all your electrical appliances before replacing the fuse. You also need to make sure the correct amp fuse wire is used.

A circuit breaker does the same job as a fuse but it does it electronically. The circuit breaker will _____ when it detects large _____.



Safety Switch / RCD

A safety switch monitors how much _____ is flowing through the _____. If the amount coming in is not the same as the amount coming _____ of the house then there must be a _____. The switch will then _____ the circuit.

Some powerboards have their own safety switch.

